

The Long-Run Trend of the Rate of Profit in the Capitalist Mode of Production

Zhihu · 2026-02-23

1. The Central Position of Profit in Capitalist Production

Capitalist production revolves around profit: the purpose of production is to obtain profit, to achieve the self-expansion of capital.

Of the two fundamental classes, the bourgeoisie, as the representative of capital, has its economic life expressed by the formula $G-W-G'$ in Capital. G and W are German abbreviations — Geld (money) and Ware (commodity). The capitalist engages in production in order to obtain a larger, expanded G' through the investment of an initial G .

The proletariat is different: it works to obtain the means of subsistence, selling its labour power as a commodity for wages, then exchanging wages for consumer goods. In this respect the working class is not very different from the small commodity producer; its formula is $W-G-W$. Note that here W does not become W' — there is no self-expansion.

Hence in capitalist society the proletariat, the great majority of the population, produces for use value, while the bourgeoisie, a small minority, produces for value. Sometimes in self-media videos we hear sayings like "the poor earn money to spend; the rich earn money to make more money, so the rich get richer." But the mover of capitalist production is not the proletariat, the largest class; it is the capitalist. Though few in number, the capitalists decide whether capitalist production expands or contracts, whether the economy booms or busts.

Since $G \rightarrow G'$ is the purpose of production, $P = (G' - G)/G$ — the rate of profit — is the key to whether the capitalist invests. Ignoring the interest rate: if the rate of profit is negative, production contracts; if positive, it expands. Taking the interest rate (I) into account: if $P - I$ is negative, production contracts; if positive, it expands. To simplify the analysis, we will ignore the interest rate from here on.

2. Value and the Rate of Profit

The bearer of G is money, so essentially G (at time T_0) $\rightarrow G'$ (at time T_1) is a process of investing some money and receiving more money. But because $G \rightarrow G'$ unfolds in time, although the face value

of G' is greater than G when the capitalist receives it, it may actually exchange for fewer commodities than the original G . There are several cases:

- If the level of productive forces is unchanged and the currency is overissued and depreciates, then at T_1 the face value of G' is larger, but it exchanges for fewer commodities than G at T_0 — production is losing money. In this case the capitalist will hoard durables or real estate rather than invest in production.
- If the level of productive forces rises and the currency is not overissued, G' at T_1 exchanges for more commodities than G at T_0 , but less than G at T_1 — the real purchasing power of the money has fallen. In this case investing is also a loss, so the capitalist would rather deposit the money in the bank.
- If the level of productive forces rises but the currency is overissued at a rate greater than the rise in productivity, then G' at T_1 has a larger face value and exchanges for more than G at T_0 ; but the increase of G' relative to G in real commodity terms is less than the increase of converting G into precious metals or real estate. Here too the capitalist loses, and will leave capitalist production if the chance arises.

Therefore, G and G' express not a change in face value but a change in value — in Marxist terms, an increase in "the mere congelation of homogeneous human labour" in the commodity. On why price is based on value, we have already discussed this in an earlier article: see "Is the Labour Theory of Value Right After All?"

The process from G to W , in which the capitalist purchases the means of production and labour power, can be expressed by C and V respectively. In the production process, labour power and the means of production combine to yield the product, whose value is the sum of the C and V' they represent. Since labour time is the sole source of value (a peculiarity of the capitalist mode of production, not a denial that machines and nature are important in production), C does not grow in the labour process, while V can become V' — the V the capitalist pays for labour power can differ from the V' that labour power creates, larger or smaller, or of course equal. Finally the capitalist sells W and receives $G(C + V')$. $G' - G = (C + V') - (C + V) = V' - V$. Marx defines V as the value of the money used to purchase labour power, and writes V' as $V + S$, so $V' - V = S$. This S is the value that labour power creates for the capitalist beyond its own value in the production process — the famous surplus value.

Thus the rate of profit can finally be expressed as $S/(C + V)$.

3. The Determinants of the Rate of Profit

Dividing both numerator and denominator of $S/(C + V)$ by V , we get $(S/V) / (C/V + 1)$. Hence the rate of profit is determined by two factors: S/V , which Marx calls the rate of surplus value, positively proportional to the rate of profit; and C/V , which Marx calls the organic composition of capital, inversely proportional to the rate of profit. That is, the higher the rate of surplus value, the higher the rate of profit; the greater the organic composition of capital, the lower the rate of profit.

How does S/V change? S is surplus value — the value the worker creates beyond the value of labour power within labour time; V is the value of labour power. Both are variable. With the development of the productive forces, V can fall, because the necessities of life become cheaper. But when the militancy of the working class strengthens, V can rise. And S and V are both accomplished within the working day: if V rises, S falls; if V falls, S can rise. If the total working time lengthens, with V unchanged S rises; if the total time shortens, with V unchanged S falls. Hence it is hard to say abstractly how the rate of surplus value S/V will change; under different social conditions and balances of class forces, this value may differ greatly.

How does C/V change? C is the value of the means of production, V the value of labour power. By the common understanding, capitalist society constantly replaces people with machines, letting one worker operate more and more machines, so C relative to V must grow ever larger. But this is only in terms of use value; in terms of value it is not necessarily so. For although the physical quantity of machinery grows, with the development of the productive forces the value it represents may fall. Hence the organic composition of capital need not increase linearly; it can also decrease.

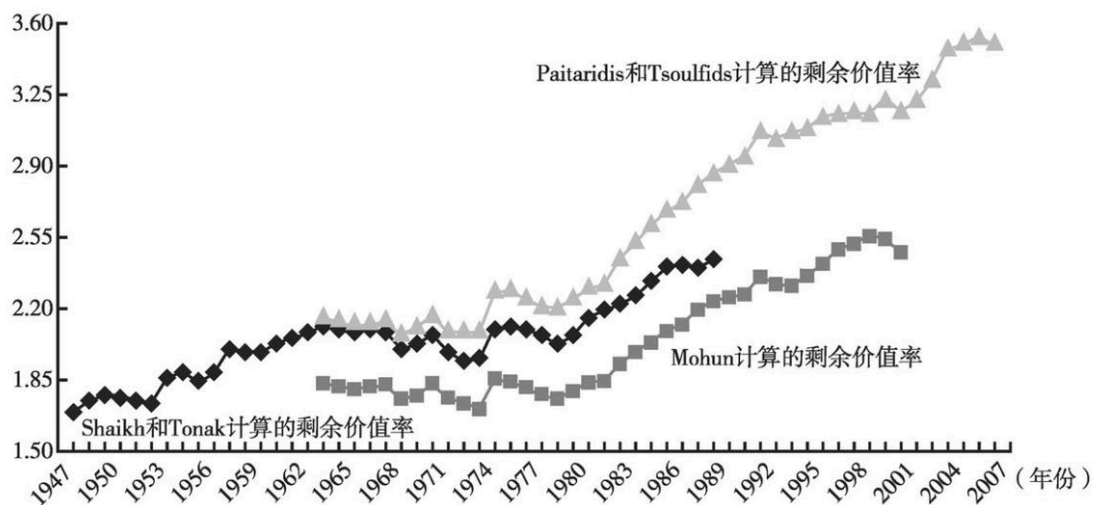
So in the rate of profit $(S/V)/(C/V + 1)$, both numerator and denominator can grow or shrink; a definite conclusion cannot be drawn by pure theory alone. In Volume III of Capital, Marx put forward the tendency of the rate of profit to fall: his reason was that the organic composition of capital tends on the whole to rise, so the rate of profit tends to fall. But he also raised many factors counteracting this tendency, such as the rise of the rate of surplus value, so he gave no definite conclusion. Later scholars have both opposed the tendency (e.g. Paul Sweezy^{[1][2]}) and...

3. Empirical Research on the Rate of Surplus Value, the Organic Composition, and the Rate of Profit

(1) The Rate of Surplus Value

The historical trend of the American rate of surplus value is shown below^[3]. After the Second World War and before the beginning of the neoliberal counter-revolution, the rate of surplus value was fairly stable; since the neoliberal restructuring began in 1980, it has risen greatly.

图片名称：美国的剩余价值率



所属图书：历史唯物论与马克思主义经济学

出版时间：2016年06月

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China's rate of surplus value, according to Qi Hao's paper^[4]:

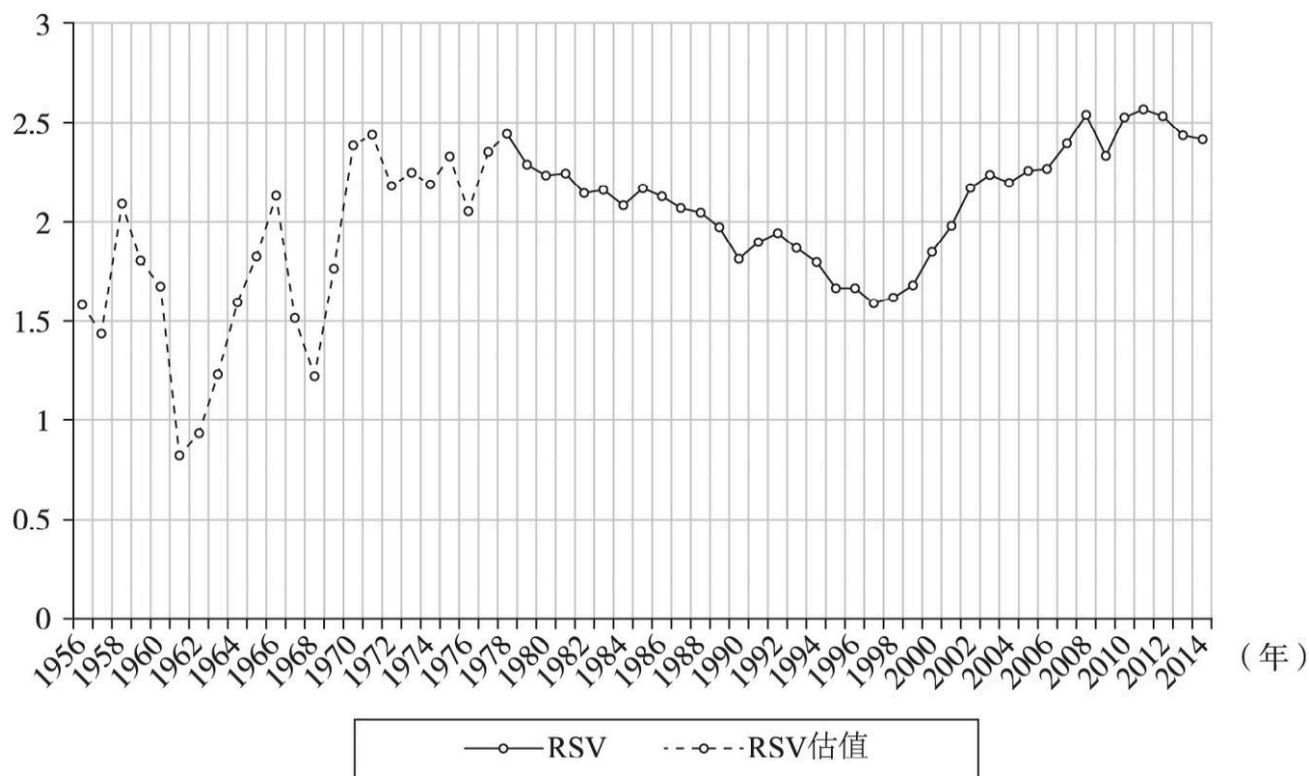


图1 1956 ~ 2014 年中国剩余价值率

注：RSV 为剩余价值率。由于 1956 ~ 1977 年估算的成分更高，因此用虚线表示以示区分。

资料来源：本文数据附录。

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(2) The Organic Composition of Capital

According to the data and charts in Shaikh's book^[5]:

184 *The Falling Rate of Profit as the Cause of Long Waves*

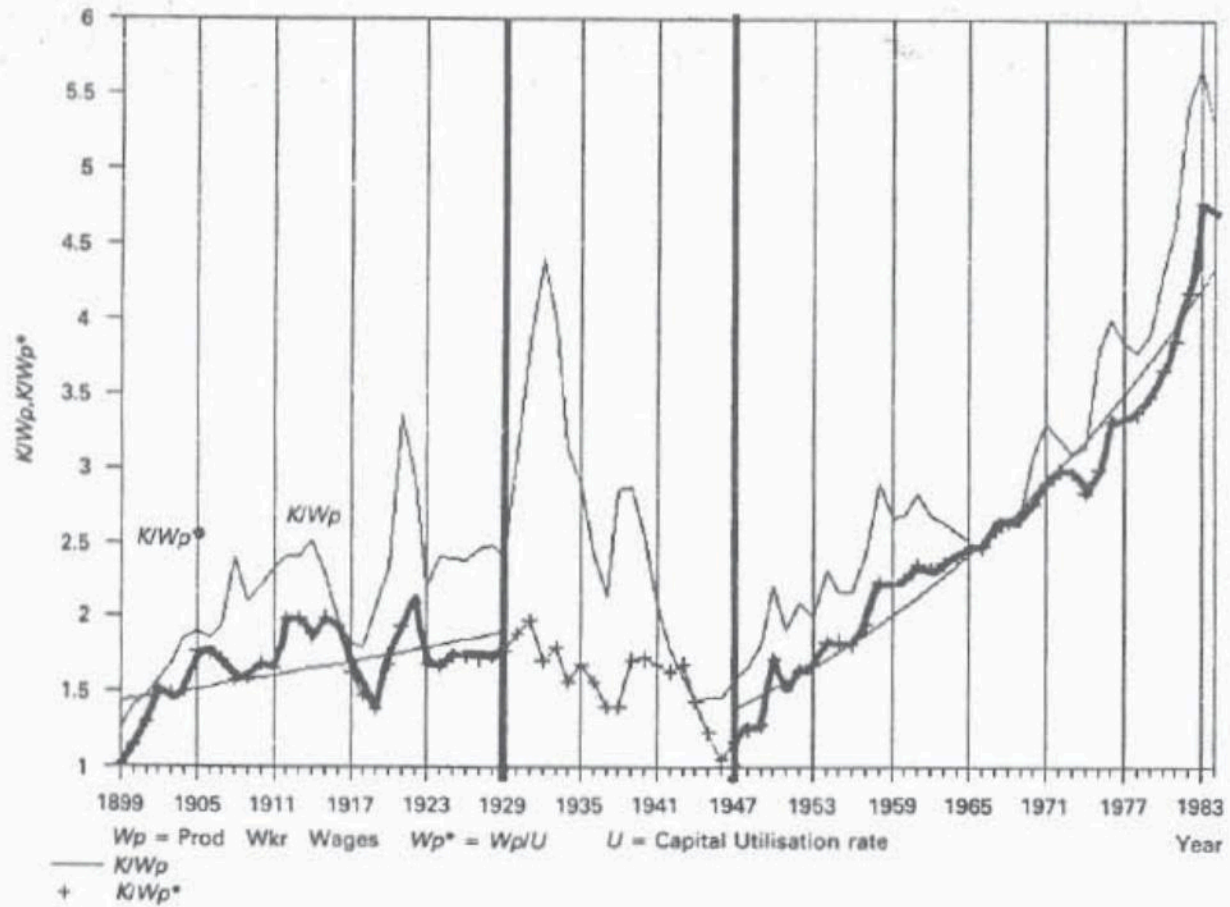


Figure 7.4 Capital/wages in manufacturing 知乎 @最最遥远的路

Figure 7.4 Capital/wages in manufacturing

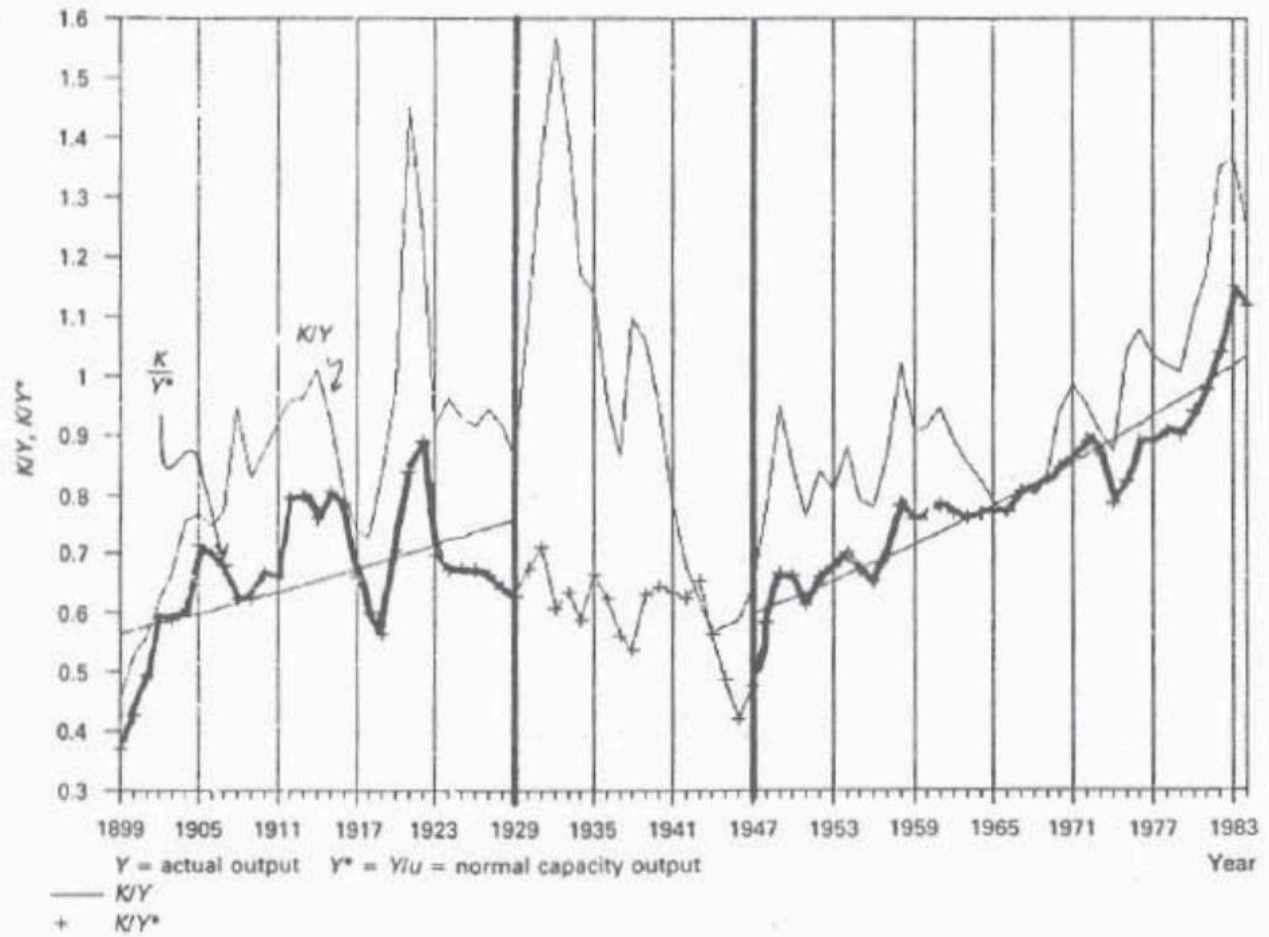
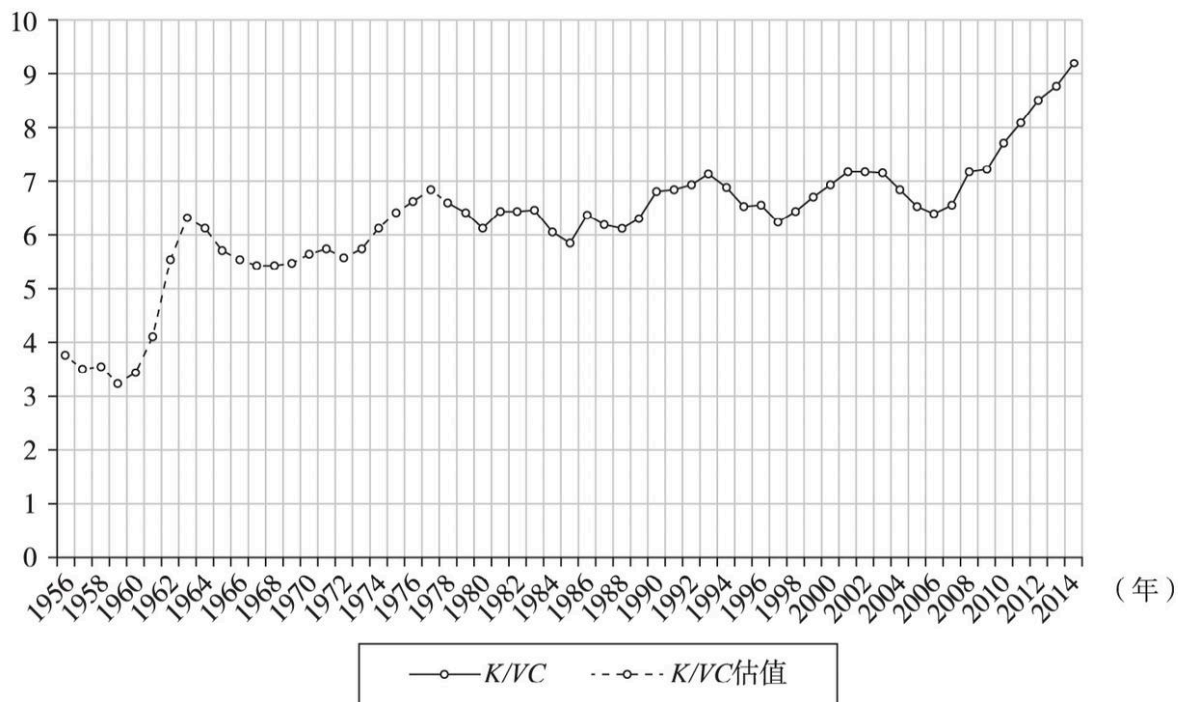


Figure 7.5 Capital/output in manufacturing 知乎@最最遥远的路

China's organic composition of capital, according to Qi Hao's paper^[6]:



注: K/VC 为资本价值构成。由于 1956 ~ 1977 年估算的成分更高, 因此用虚线表示以示区分。

图 3 1956 ~ 2014 年资本价值构成

资料来源: 本文数据附录。

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(3) The Rate of Profit

The chart below^[7] shows the average rate of profit: from the post-war period until the neoliberal counter-revolution it was basically declining; after 1980 it rebounded somewhat, then hovered at a low level.

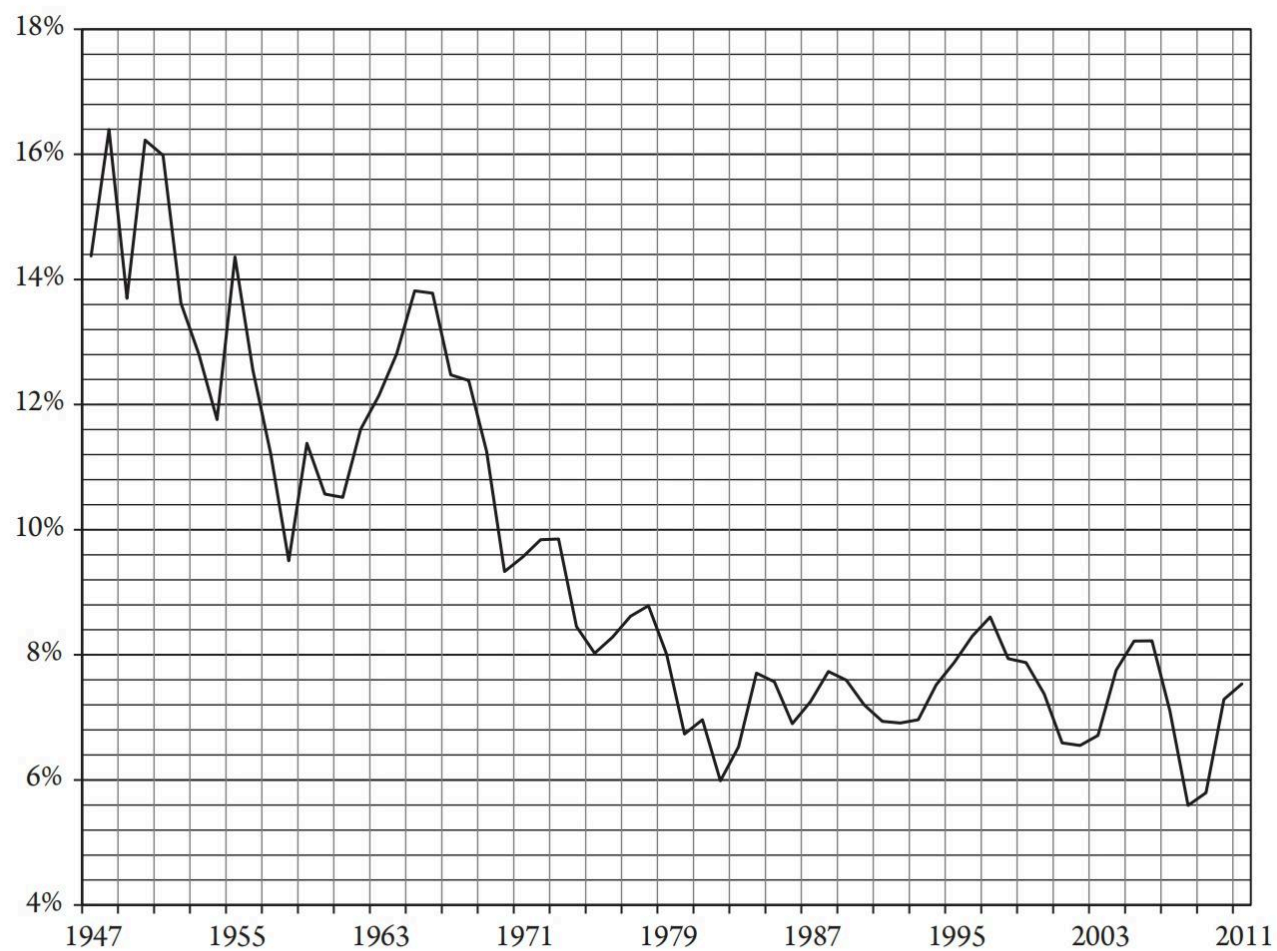
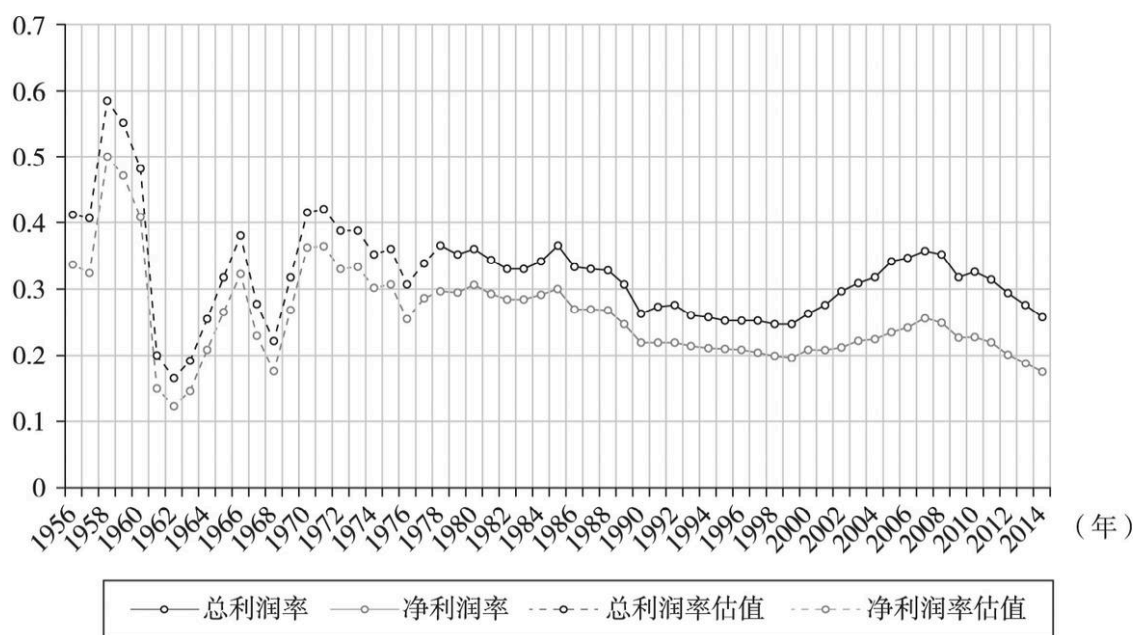


Figure 2.11 US Corporate Rate of Profit 1947–2011

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According to Qi Hao's paper^[8]:



注：总利润率（gross rate of profit, GRP），净利润率（net rate of profit, NRP）。由于1956~1977年估算的成分更高，因此用虚线表示以示区分。

图4 1956~2014年中国的总利润率和净利润率

资料来源：本文数据附录。

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(4) The Causes of Changes in the Rate of Profit

According to the analysis in Li Chong's paper "Verification and Analysis of Marx's Law of the Tendency of the General Rate of Profit to Fall," the organic composition of capital affects the rate of profit, but less than the rate of surplus value — a relationship that holds both before and after the neoliberal reforms:

From 1948 to 1982, judging by the change in the organic composition of capital, it rose from 4.31 in 1948 to 5.01 in 1982. In this period, the correlation coefficient between the general rate of profit and the organic composition of capital was -0.30 — a certain negative correlation. The organic composition rose 16.24% while the general rate of profit fell 27.88%. As for the rate of surplus value, it fell from 54.85% to 44.81%. In this period, the correlation coefficient between the general rate of profit and the rate of surplus value was 0.79 — a fairly strict correlation. The rate of surplus value fell 10.04% while the general rate of profit fell 27.88%. This means that in this period the tendency of the general rate of profit to fall was caused by the rise of the organic composition and the fall of the rate of surplus value, but the rate of surplus value had the greater influence. From 1982 to 2006, the organic composition fell from 5.01 to 4.63. In this period, the correlation coefficient between the general rate of profit and the organic composition was -0.24 — a certain negative correlation. The organic composition fell 7.58% while the general rate of profit rose 42.42%. As for the rate of surplus value, it

rose from 44.81% to 59.78%. In this period, the correlation coefficient between the general rate of profit and the rate of surplus value was 0.90 — a high correlation. The rate of surplus value rose 10.04% while the general rate of profit rose 33.41%. This means that in this period the tendency of the general rate of profit to rise was caused by the fall of the organic composition and the rise of the rate of surplus value, but the rate of surplus value had the greater influence^[9].

Qi Hao's paper^[10]:

In the first stage (1978–1990), the rate of surplus value fell while the value composition of capital rose; in this stage the rate of surplus value was the main factor in the change of the net profit rate. The second stage (1990–1999) continued the pattern of the first: the rate of surplus value remained the main driver. Although the value composition of capital fell slightly, its positive effect on net profit rate growth was just offset by the negative effect of surplus extraction by the non-productive sectors. In the third stage (1999–2007), the sharp rise of the rate of surplus value reversed the declining trend of the net profit rate, giving the economy its fastest net-profit-rate growth since reform and opening up. In this stage the value composition of capital and the influence of the non-productive sectors were of secondary importance. In the fourth stage (2007–2014), because the value composition of capital rose rapidly, the net profit rate fell sharply; the stagnant rate of surplus value could not offset the influence of the value composition, and the rising share of the non-productive sectors also pressed the net profit rate down.

The rise of the organic composition of capital comes from technical progress. Technical progress means replacing people with machines, but at first technical progress may sharply depreciate the existing constant capital, lowering the organic composition. As replacement with new-technology equipment begins, the organic composition rises again; in the very early stage, investment bubbles easily appear — like the internet investment boom of the late 1990s. But the rising organic composition presses the rate of profit down, slowing investment, which slows the growth of the organic composition — until the next technological revolution arrives and the cycle begins again. Thus technical progress and investment are two opposing factors driving changes in the organic composition.

The rise of the rate of surplus value depends on supply and demand in the labour market and the balance of class forces: with an abundant industrial reserve army, the rate of surplus value can stay high. But when capital accumulation becomes too large and the reserve army is relatively depleted, the militancy of the working class strengthens, wages tend to rise, and the bourgeoisie can reduce investment, increase unemployment, and finally push wages back down. Hence the fall of the rate of surplus value leads to its own rise.

Cockshott's paper^[11]:

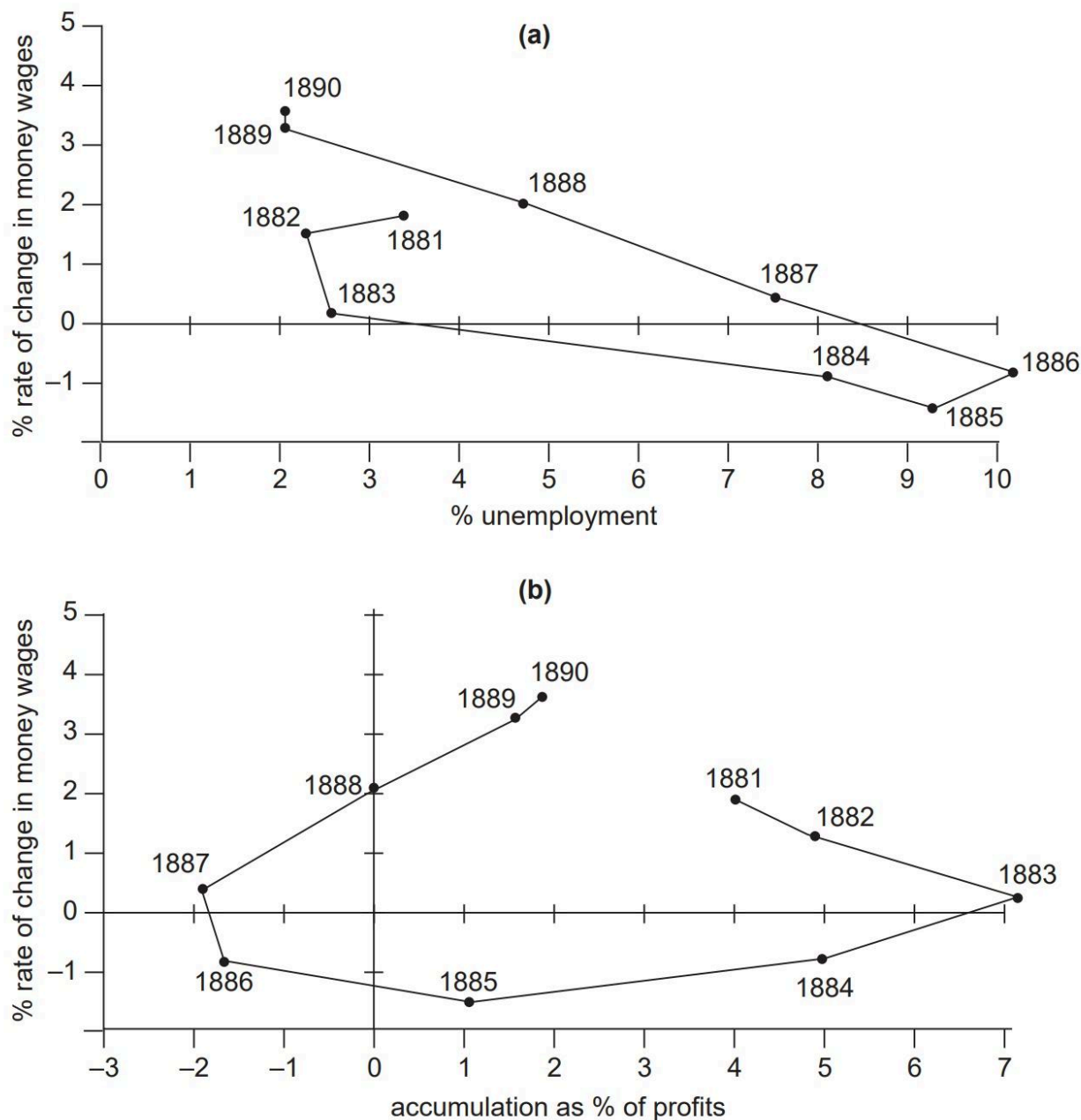


Figure 2 The relationship between accumulation, unemployment and rate of change of wages

Source: Reproduced from Cockshott (1977).

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But these trends are all short-term, presenting as fluctuations. Over a span of decades, how is the long-run trend of the capitalist rate of profit determined?

4. The Long-Run Trend of the Rate of Profit

In this part, I want to introduce a model of the long-run trend of the rate of profit proposed by Cockshott^[12], which has great significance for understanding the evolution of the world-system.

Let $G = d\lambda/dt$ be the annual growth rate of the labour force. Let α be the accumulated share of net surplus value. Assume the rate of profit is stable, and denote this stable rate of profit as p^* . Obviously, for the rate of profit to be stable, the following relation must hold: $p^* = G/\alpha$ (1). Why? Suppose the rate of profit is s/C ; then $\alpha p^* = \alpha s/C$ is the annual growth rate of the capital stock (s being surplus value). $\alpha p^* = \alpha s/C = dC/dt$. But we define p^* as the stable rate of profit, at which $G = dC/dt$ — the change in capital corresponds to the change in population. $\alpha p^* = \alpha s/C = dC/dt = G$, hence $p^* = G/\alpha$. Let S denote the total surplus, d the annual depreciation rate (as a share of capital), $P = S/C$ the gross rate of profit, and A the share of gross surplus used for gross accumulation. Now we can modify formula (1) to get the stable gross rate of profit: $P^* = (G + d)/A$ (2). The rapid development of technical change tends to raise the stable rate of profit, because it slows the process of accumulation measured in real value — that is, accumulation measured in years of labour represented by the capital stock. Let t be the growth rate of labour productivity from technical change; our final equation is: $P^* = (G + d + t)/A$ (3). P^* can be regarded as the dynamic attractor of the rate of profit. If the variables A , G , d , and t remain unchanged, the actual gross rate of profit P should asymptotically approach P^* .

Reading equation (3): the long-run trend of the rate of profit is proportional to the population growth rate, the capital depreciation rate, and the rate of technical progress, and inversely proportional to the accumulation rate.

- When technical progress stagnates, d and t basically stabilise; if the labour force falls, the accumulation rate must fall in step to maintain the rate of profit — the general situation of the developed countries with deep ageing: population falling, yet unemployment not low.
- When the labour force is stagnant, if technical progress suffices to offset population change, the rate of profit can be maintained with the accumulation rate unchanged — the situation the developed countries, including China, hope for, though technical progress is hard to sustain long.
- When the labour force grows and technical progress grows in step, capital can achieve rapid accumulation — the situation of late-developing countries in the period of rapid industrialisation: post-war Europe, Japan, and China of the last thirty years.

5. Empirical Verification of the Long-Run Trend Formula

Cockshott's paper^[13] verifies the validity of equation (3) with data; one can see that the actual rate of profit lags P^* by a few years:

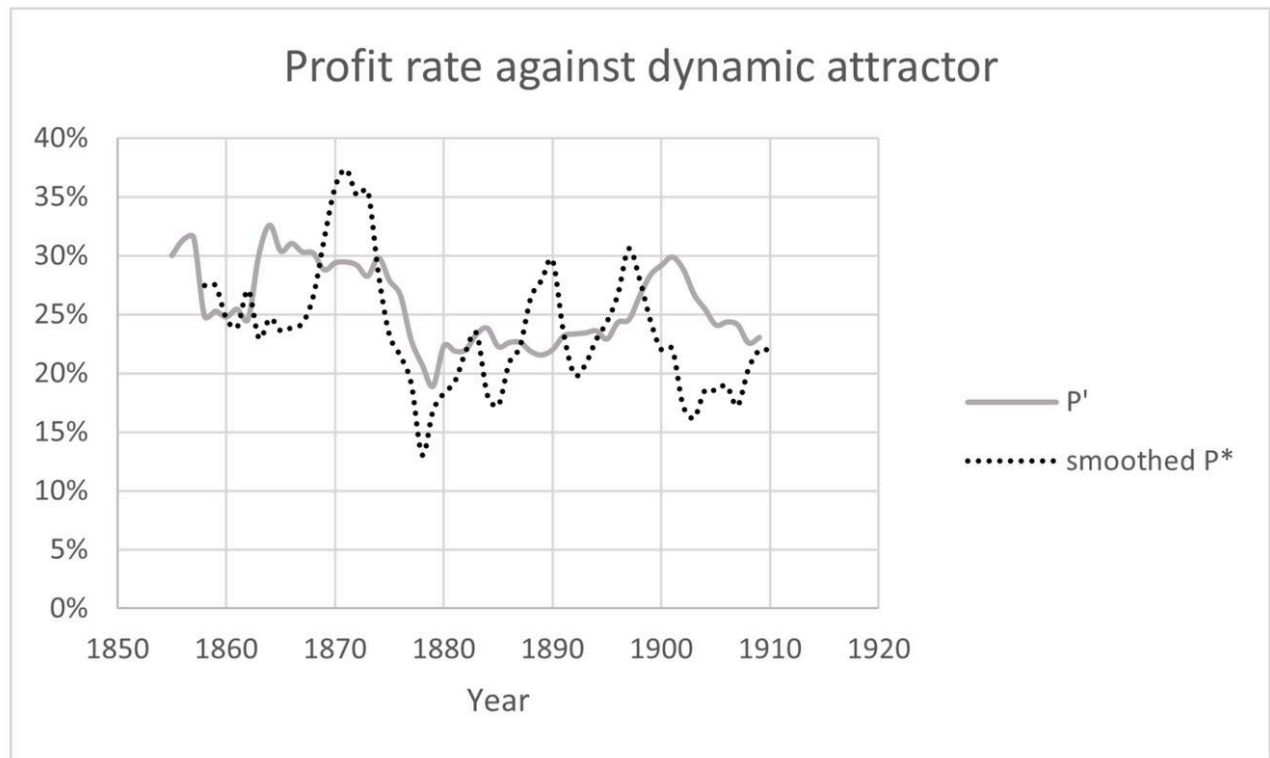
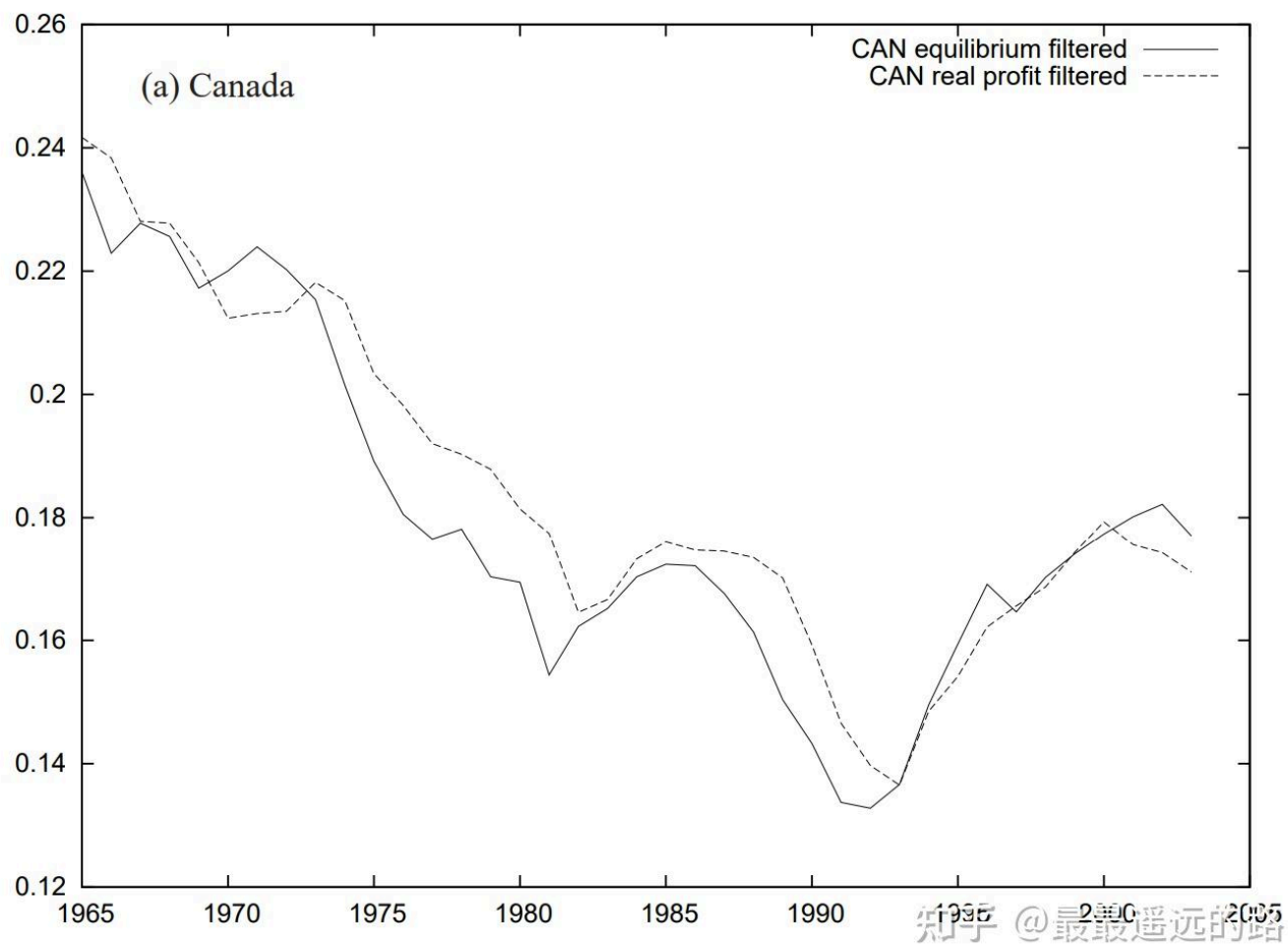


Fig. 10: P' is the gross rate of profit and smoothed P^* is the 4 year average of the dynamic profit rate attractor. All data calculated from *A Millennium of Economic Data*.

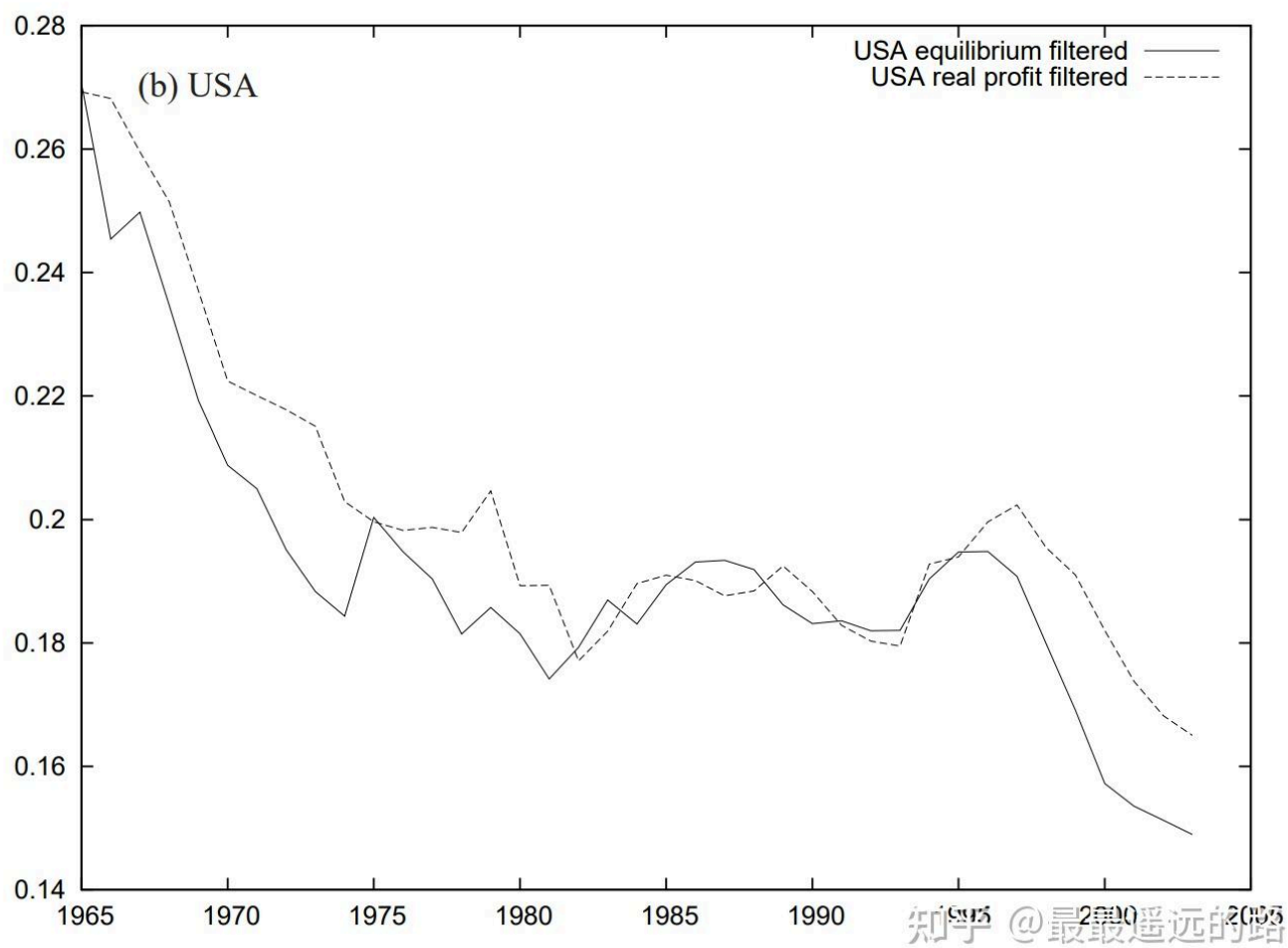
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In another paper^[14]:

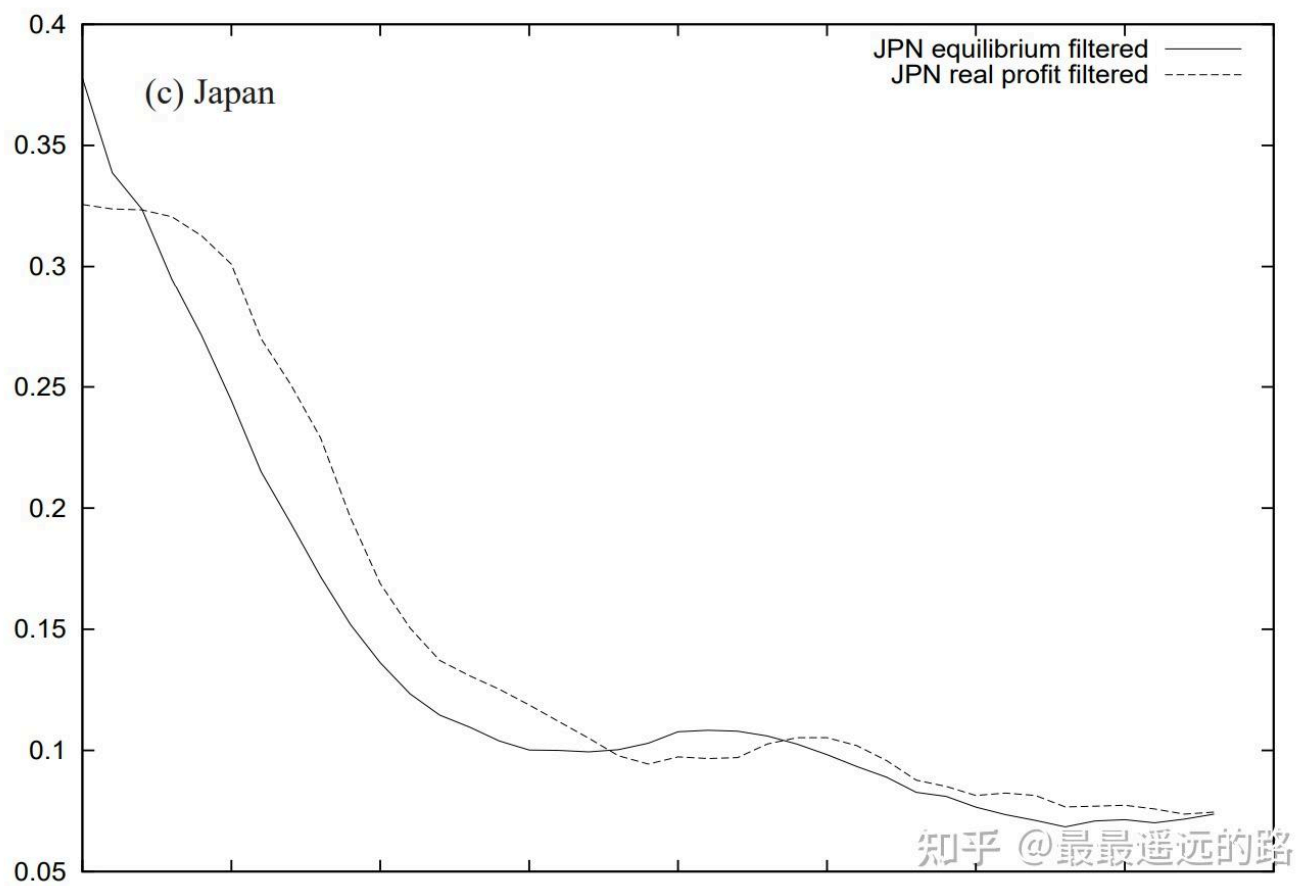
Canadian data verification:



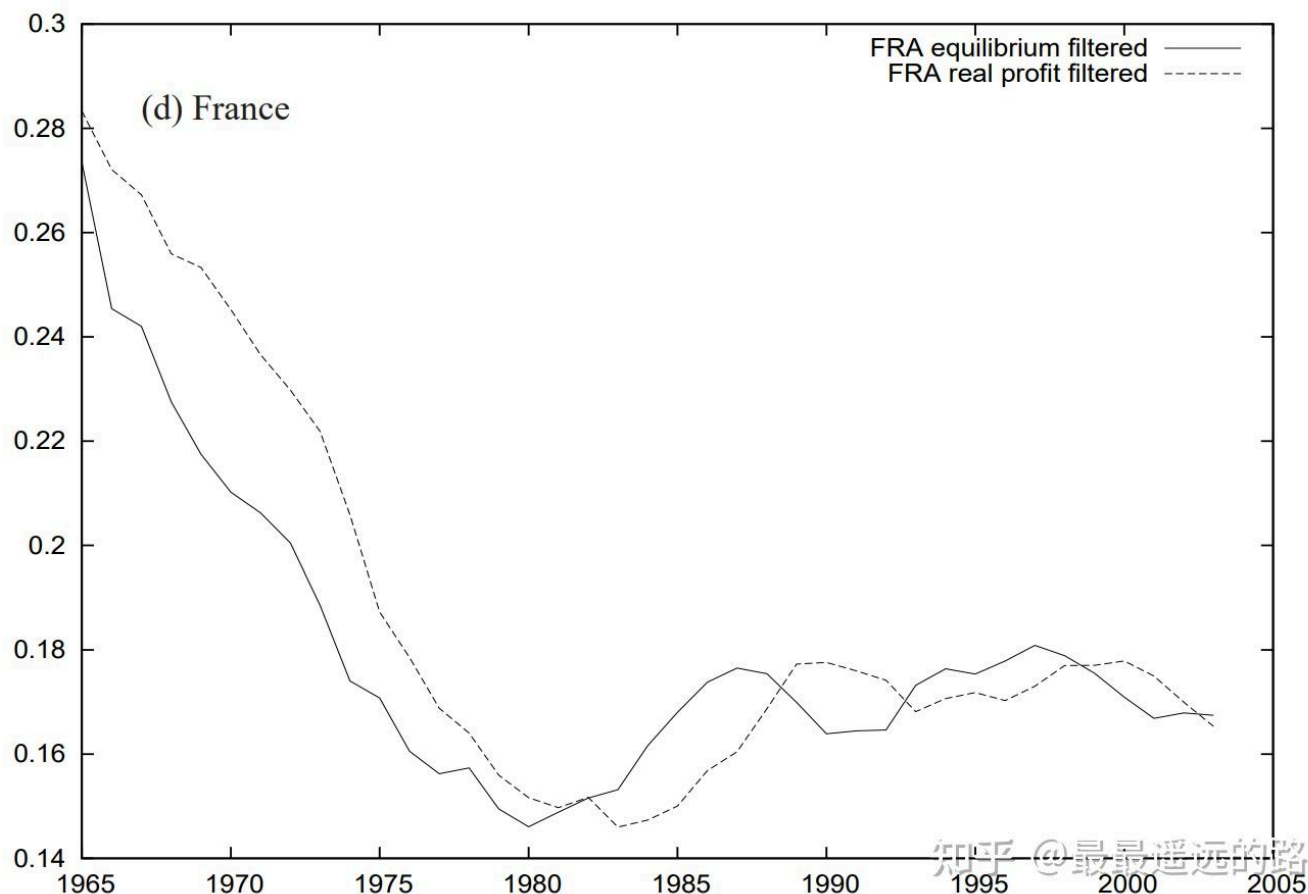
US data verification:



Japanese data:



French data:



6. The Long-Run Trend of the Rate of Profit in the Capitalist World-System

In the current world-system, large numbers of peasants still join the proletariat every year and the labour force keeps growing, so capital accumulation continues at a very high rate in the periphery. This is, of course, the result of capitalism's development from national industrial capitalism, facing its accumulation crisis, through neoliberal reforms, into transnational production capitalism.

But this process cannot be maintained forever. When the rural surplus labour of the globe has basically been absorbed into capitalist production, and the world has basically urbanised, the growth of the labour force (G in equation 3) can no longer come from peasants entering the cities, but only from the population's own growth.

The development of capitalism brings proletarian impoverishment, class ossification, feminism, the dissolution of kinship ties, and other factors that push the birth rate down. The only regions still maintaining high fertility are South Asia and sub-Saharan Africa; in South Asia, apart from Pakistan, fertility has already fallen near the replacement level. People used to think Africa's high fertility could last a long time, but recent developments show that African fertility may well fall rapidly — the so-called African population explosion will not occur.

The Economist's 2023 article "The world's peak population may be smaller than expected: fresh evidence suggests African birth rates are falling fast"^[15]:

In 1972, the think tank Club of Rome published its hugely influential book *The Limits to Growth*, warning that consumption and population growth would cause economic collapse. Now the club says a population explosion may never happen: it estimates that sub-Saharan Africa's population peak may come as early as 2060, 40 years earlier than the UN's forecast. The UN's population projections are widely regarded as the most authoritative. Its latest report last year projects sub-Saharan Africa's population far lower than its forecast a decade ago. Nigeria, Africa's most populous country with about 213 million people, has seen its UN projection for 2060 cut by more than 100 million (to about 429 million). By 2100, Nigeria's population is expected to reach about 550 million, 350 million fewer than forecast a decade ago. However, even the UN's latest projections may fail to keep pace with the rapid fall of fertility (the average number of children a woman is expected to bear). Some striking recent studies show this trend. The most striking is Nigeria: an unverified 2021 survey found its fertility had fallen from 5.8 five years earlier to 4.6. Another survey supported by USAID appears to confirm roughly the same, finding a fertility rate of 4.8 in 2021, down from 6.1 in 2010. If these findings are correct, they indicate a rate of birth-rate decline similar to parts of Asia, where population growth slowed sharply in a process often called the demographic transition^[16].

The Lancet's 2020 paper "Fertility, mortality, migration, and population scenarios for 195 countries and territories from 2017 to 2100: a forecasting analysis for the Global Burden of Disease Study"^[17] projects the global population peaking in 2064 at 9.73 billion. Since the working-age population peaks about ten years before the total population (China: 2012 and 2022; Japan: 1995 and 2008; India: expected 2050 and 2060 respectively), the global working-age population will peak around 2054.

Combining mortality, fertility, and migration, we forecast that the global population will peak in 2064 at 9.73 billion (95% UI 8.84–10.9 billion) in the reference scenario, then decline to 8.79 billion (6.83–11.8 billion) in 2100.

After the 2050s, as the last rural labour forces are absorbed into capitalist production (partly migrating to other regions to supplement their labour, partly entering local cities to work, including in foreign-owned factories), the scale of the labour force under the global capitalist mode of production will fall into contraction no later than the 2060s. According to formula (3), when G begins to decline, the long-run equilibrium rate of profit also declines; without a great explosive technical progress, the rate of profit can finally be restored only by lowering the accumulation rate. That is, capitalism will fall into stagnation: vast quantities of surplus value will no longer be accumulated for expanded reproduction and the formation of capital, but will be consumed by the bourgeoisie or spent on non-productive expenditure such as military budgets and war. Weak wage growth and

widespread unemployment will become global phenomena. At that point, capitalism's historical task will have been accomplished, its vitality exhausted — and the era of social revolution will have arrived.

Notes

[1] "Recent Criticisms of Marx's Law of the Falling Rate of Profit" (Rosdolsky): In the whole of Marx's system, few fundamental principles have been attacked with such unanimity by academic and non-academic critics alike as his law of the tendency of the rate of profit to fall. Yet the arguments of Marx's critics may be the least satisfactory. Two writers now focusing on this law may serve as examples: Joan Robinson and the American Marxist Paul Sweezy. As early critics of this law, Robinson and Sweezy held that Marx committed a fundamental methodological error. Marx is said to have expounded his law of the falling rate of profit on the assumption of an unchanged rate of surplus value, arbitrarily separating the factors lowering the rate of profit from those raising it, so as to derive his law of the falling rate of profit from the former and the "counteracting influences" from the latter. Or, as the early supporter of this criticism, von Bortkiewicz, put it: "The error in Marx's brief proof of his law of the falling rate of profit is that he ignores the quantitative relationship between labour productivity and the rate of surplus value. Marx regards the rate of surplus value as an irrelevant factor. The fallacy usually arising from this method of separating related factors into irrelevant ones can be seen from the following simple example. Suppose a positive value a , related to other positive values b and c through the function $a = b/c$."

中文原文：《近来对马克思利润率下降规律的批评》在马克思的全部体系中，很少能有象他的利润率趋向下降规律这样一条基本原理遭到学院派和非学院派如此一致的批驳。然而，马克思的批评者对此所持的论据可能是最不能令人满意的。现在关注于马克思这一规律的两作家可以作为例证。他们是琼·罗宾逊[1]和美国的马克思主义者保罗·斯威齐。[2] 作为对马克思这一规律的早期批评，罗宾逊和斯威齐认为，马克思犯了一个根本的在方法论方面前后矛盾的错误。据说马克思是在假定剩余价值率不变的情况下阐述其利润率下降规律的，他武断地把降低利润率的因素与提高利润率的因素割裂开来，以便从降低利润率的因素中导出他的利润率下降规律，从提高利润率的因素中导出“起抵销作用的影响”。或如这种批评意见的早期支持者冯·鲍特凯维茨所说的那样：“马克思对其利润率下降规律所提供的简要证明中的错误在于，他忽略了劳动生产率和剩余价值率之间的数量关系。马克思认为剩余价值率是个不相关的因素。这种把相关因素割裂为不相关因素的方法通常导致的谬误能够从下面简单的例子中看出来。假设正值 a ，它与其他正值 b 和 c 通过函数 $a=b/c$ 相关联。

<https://www.marxists.org/chinese/rosdolsky/1969/28.htm>

[2] "Kliman's Marxist Analysis of Economic Crisis — a re-proof of the law of the tendency of the rate of profit to fall"

中文原文：《克莱曼对经济危机的马克思主义分析——利润率下降趋势规律的再证明》

<http://www.ddjjyj.com/CN/abstract/abstract528.shtml>

[3] "The rate of surplus value in the United States"

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https://www.pishu.com.cn/skwx_ps/multimedia/ImageDetail?SiteID=14&type=Picture&ID=13082041&ContentType=MultimediaImageContentType&isHost=null

[4] "The Change in the Rate of Surplus Value and China's Economic New Normal: based on the method distinguishing productive and unproductive labour" (Qi Hao)

中文原文：《剩余价值率的变动与中国经济新常态：基于区分生产劳动与非生产劳动的方法》

<https://www.lishiyushehui.cn/statics//article/attachment/202504/rj4FJkbjxxE6R4vI.pdf>

[5] Anwar Shaikh, "The Falling Rate of Profit as the Cause of Long Waves: Theory and Empirical Evidence"

中文原文: The Falling Rate of Profit as the Cause of Long Waves: Theory and Empirical Ev

<http://gesd.free.fr/shaikh92w.pdf>

[6] "The Change in the Rate of Surplus Value and China's Economic New Normal" (Qi Hao)

中文原文: 《剩余价值率的变动与中国经济新常态: 基于 区分生产劳动与非生产劳动的方法》

<https://www.lishiyushehui.cn/statics//article/attachment/202504/rj4FJkbjxxE6R4vI.pdf>

[7] Anwar Shaikh, Capitalism: Competition, Conflict, Crises

中文原文: 《Capitalism : competition, conflict, crises》66页

<https://archive.org/details/capitalismcompetoooshai>

[8] "The Change in the Rate of Surplus Value and China's Economic New Normal" (Qi Hao)

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<https://www.lishiyushehui.cn/statics//article/attachment/202504/rj4FJkbjxxE6R4vI.pdf>

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This article was originally published on Zhihu (in Chinese): <https://zhuanlan.zhihu.com/p/2009217555714306113> Author: 最最遥远的路 · Zhihu profile: <https://www.zhihu.com/people/zui-zui-yao-yuan-de-lu-32/posts>

本文原发表于知乎专栏/回答（中文）： <https://zhuanlan.zhihu.com/p/2009217555714306113>；作者：最最遥远的路： <https://www.zhihu.com/people/zui-zui-yao-yuan-de-lu-32/posts>